Vector-Shield: Weekly Project Progress Report

Supervisor: Dr. Lekshmi C. R. | **AIDS Team 14:** Vignesh V S, Karthick Ruban T, Sanjay Senthil, Hari Prakash V
Status: Completed Adaptive Centroid Evolution, Multimodal Vision Shield, 1,200 Benchmark, & MDPI Math Outline

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Respected Ma'am,

Vector-Shield: Weekly Project Progress Update (AIDS Team 14)
Supervisor: Dr. Lekshmi C. R.

Wishing you and your family a very Happy and Prosperous Diwali in advance! ■■

As per your review guidance, we prioritized the first two developmental directions and have successfully completed them this week. We have also pushed all updates to GitHub and refreshed our live Streamlit deployment.

1. Priority 1 Completed: Adaptive Centroid Updating & Mathematical Framework
Streaming Centroid Evolution (EMA):* Implemented dynamic Exponential Moving Average updates (c_{t+1}) in 'src/lof_interceptor.py'. The enterprise knowledge centroid now evolves in real-time as verified corporate documents are ingested, eliminating static drift.
Adversarial Drift Gate:* Formulated a strict mathematical containment gate: only vectors with LOF_scaled <= 1.30 are permitted to update the centroid, mathematically preventing centroid poisoning / creep attacks.
Live Telemetry:* Integrated real-time tracking of cumulative L2 drift distance, condition bounds, and dynamic cluster radius (tau_t) in the pipeline.

2. Priority 2 Completed: Multimodal Prompt Injection Detection (Vision Shield)
Multimodal Interception Engine:* Implemented 'src/multimodal_shield.py' to screen visual documents, scanned PDF invoices, receipts, and charts.
Cross-Modal Topographic Incoherence (CMTI):* Formulated CMTI(I, T) = 1 - cos(e_vision, e_text) to catch typographic prompt overrides hidden in scanned document layouts (e.g., hidden micro-text or conflicting invoice commands).
Dashboard Integration:* Added a dedicated '■■ Multimodal Vision Shield' tab in our live Streamlit app with interactive presets for clean vs. typographic override invoices.

Live Deployment & GitHub Repository:
Live Streamlit App (Updated with new modules):* https://vectorshield.streamlit.app/
GitHub Repository (Latest codebase pushed):* https://github.com/VigneshVS2005/Vectorshield

Work in Progress for Next Weekend:
Experimental Benchmark & Mathematical Formulation:* We are currently actively working on the complete dataset scaling (expanding to multi-domain samples with full train/val/test splits) and the mathematical proofs for MDPI Mathematics. We will finalize these and share the detailed report by next weekend.
Next Set of Priorities:* We are also very keen to take up and implement the next 3 developmental priorities from our roadmap by next weekend.

Ma'am, since it is Diwali festival time, there is absolutely no hurry to review this right now. We wanted to keep you regularly updated with our weekly progress. Please review it at your convenience and share your valuable suggestions and feedback before we move forward.

Wishing you a joyful Diwali celebration once again, Ma'am! ■

Respectfully,
Team 14 (Vignesh, Karthick, Sanjay, Hari)